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Published in:
RE:SOURCE 2023 The 10th International Conference on the Histories of Media Art, Science and Technology 13 16 September 2023 Venice, Italy. PROCEEDINGS

Publication date:
2024

Document version:
Accepted manuscript

Citation for pulished version (APA):
Jørgensen, J. (2024). Convulsive Automation. Notes on the Resurgence of Surrealist Aesthetics Within Soft Robotics. In *RE:SOURCE 2023 The 10th International Conference on the Histories of Media Art, Science and Technology 13 16 September 2023 Venice, Italy. PROCEEDINGS* (pp. 293-296)
https://drive.google.com/file/d/1-tTEKj5ZVbrU3Btq_FZu9orn1iaf_L5x/view

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Convulsive Automation

Notes on the Resurgence of Surrealist Aesthetics Within Soft Robotics

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ABSTRACT

Surrealism and robotics share a connection through the concept of "the uncanny". Art historians have, for instance, remarked on the persistent presence of archetypically uncanny motifs, including automatons, dolls, and prostheses, within surrealist art and research on social robots frequently cite the concept of the "uncanny valley" to explain findings of human-robot interaction experiments. In this paper, however, I seek to construct a different pathway of connections between surrealism and robotics, focusing on the recently emerged field of soft robotics. Soft robotics designates a radically novel class of robotics technologies and a line of research that eschews the traditional ambitions of robotics research (e.g. the quest for precision) in favor of leveraging the functional benefits that come with constructing bioinspired machines from pliable materials with mechanical similarities to organic tissue. Albeit more often inspired by the anatomy of animals than that of humans, soft robots, arguable, sit uneasily between the perceptual categories of inanimate objects and living beings, as well as those of the intimately familiar and the morbidly strange. Soft robotic designs are often composed of silicone modules and structures with simple abstract geometrical shapes yet possess smooth surfaces with skin-like haptic qualities. During operation, they bulge out and assume highly biomorphic forms, they are able to perform smooth movements (often by means of pneumatic actuation), and diligently adapt to their surroundings. They can deform, alter their shape, or even go shapeless and dynamically change back and forth from being hard to being soft – operations and procedures that figure prominently within surrealist visuality and the surrealist imaginary. Drawing on selected surrealist artworks and theoretical writings as well as art historical accounts of surrealism, this paper presents an argument that a surrealist aesthetic can be seen to reverberate throughout soft robotics. Furthermore, it seeks to interrogate how soft robotics activates categories and themes of surrealist art practice (beyond "the uncanny"). Lastly, the question is raised as to what explanatory force surrealist concepts and ideas, notably Breton's concept of "convulsive beauty", might have for understanding and contextualizing soft robotics as an emergent cultural phenomenon – are their affinities merely a surface phenomenon or could they run deeper?

CCS CONCEPTS

• Human-centered computing~Human computer interaction (HCI)
•Applied computing~Arts and humanities

KEYWORDS

Soft robotics, surrealism, aesthetics, convulsive beauty, human-robot interaction

ACM Reference format:

Jonas Jørgensen. 2024. Convulsive Automation: Notes on The Resurgence of Surrealist Aesthetics within Soft Robotics. In *Proceedings of The 10th International Conference on the Histories of Media Art, Science and Technology (MediaArhistories'23). September, 2023, Venice, Italy. ACM, New York, NY, USA, 6 pages.* https://doi.org/10.1145/1234567890

1 Introduction

Surrealism and the field of robotics share a connection through the central surrealist practices and techniques of automatism and automatic writing and their associated ideals of being machine-like in performance to achieve unhindered and ego-less creation. Likewise, art historians have remarked on the persistent presence of archetypically uncanny motifs, including automatons, dolls, and prostheses, within surrealist art and social robotics frequently cite Masahiro Mori's concept of the "uncanny valley" to elucidate findings of human-robot interaction experiments [22]. In the present text, however, I seek to construct a different pathway of connections between surrealism and robotics. More specifically, my interest lies in interrogating affinities with the recently emerged field of soft robotics, a novel subarea of technical robotics research that endeavors to construct robots from pliable and elastic materials such as silicones for functional gains. The work is part of a larger effort to rethink and recontextualize soft robotics in relation to historical art traditions and art practices to

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MediaArhistories'23, September, 2023, Venice, Italy
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construct a cultural genealogy of their aesthetics [10–14]. The wider interest of the paper is thus to use surrealism as a lens for interrogating formal and material aspects of soft robots and their design aesthetics and to shed light on the aesthetic reception of soft robots in human-robot interaction. I will argue that facets of surrealist aesthetics reverberate throughout soft robotics and that soft research robots activate visual motifs and central thematic interests of surrealist art. Furthermore, I will seek to illustrate how concepts from surrealist theory may hold relevance for understanding and contextualizing soft robotics as an emergent cultural and material-semiotic phenomenon and for elucidating.

2 Soft Robotics and "Surrealist Style"

Soft robotics research is founded on the simple idea of constructing robots, not from metal or plastic, but from pliable materials such as silicone rubbers with mechanical similarities to soft organic tissue. Within technical robotics research, this shift is motivated by the observation that soft body parts are essential to the ways in which animals easily and adeptly can accomplish sophisticated mechanical tasks [21]. In addition, the technical advantages of soft robots over traditional robots include an increased adaptivity and better maneuverability in unstructured environments, simplified control, and increased energy efficiency [26]. Due to their passive compliance, soft robots are also considered inherently safer for humans to interact with and have been proposed for applications that require close physical contact, including collaborative work, care, and rehabilitation [25]. The research field of soft robotics eschews several of the traditional ambitions and normative criteria of robotics research. For instance, the quest for speed and precision is to some extent sidestepped in favor of leveraging adaptivity, simplification of control, and energy efficiency from constructing machines that are soft. Moreover, the change to soft materiality endows soft robot it with a different set of affordances for interaction with humans and, as will become evident in the following, entails a novel and distinct robot aesthetic. This aesthetic departs from perceptions of robots as clunky inhuman machines within the cultural imaginary [7] and instead brings the field of robotics into contact with different aesthetic traditions, including surrealist art.

The lack of a unified visual style within surrealist painting has frequently been lamented by modernist scholars. In fact, the only solution the modernists found to this problem was William Rubin's idea to bifurcate the movement into two separate branches – an "automatist/abstract" and an "academic/illusionist" [24]. This problematic of "surrealist style" of course creates difficulty in arriving at a unified and general description of the surrealist visual aesthetic especially within the privileged modernist medium of painting. However, if we, following subsequent scholarship, consider the visual and textual production of the surrealists as a whole, including, e.g., photography and print media, shared formal traits and interests obviously exists across surrealist art. Interestingly, a number of these recur in the specific aesthetic characteristic of contemporary soft robotics technology.

In the following, I will seek to introduce these overlaps through juxtapositions of select surrealist artworks and examples of soft robots and soft robotics research projects within technical research to highlight parallels between these otherwise radically disparate fields of practice. As a note of clarification, it is necessary to mention here that the selective readings proposed of specific surrealist works are anchored primarily in the specific understanding of surrealism emergent within the art historical and art critical writings of the members of October group. Most notably the paper draws on Rosalind Krauss' postformalist work on "dissident surrealism" and surrealist photography. Hence, unlike some parts of recent surrealism scholarship [6,16,23], which affirm a continuation of Surrealism in the 1950s, 1960s, and 1970s, my focus is restricted to surrealist artworks and concepts hailing from the inter-war period.

3 Biomorphism and Othered Animals

A first and perhaps most obvious point of entry to exploring potential overlaps between surrealism and soft robotics is through the shared use of biomorphic shapes. The concept of the *biomorph* was first used by British anthropologist and ethnologist Alfred Cort Haddon to describe "representation[s] of anything living" and subsequently came to denote elements within painting, sculpture or design that reference organic life forms [8]. When Alfred H. Barr referred to *biomorphism* in "Cubism and Abstract Art" (1936), he thus used this term to describe a trend of "curvilinear," "decorative," and "romantic" forms in abstract art [1].



Figure 1: Left: Hans Bellmer, *The Doll* (*La poupée*), 1949, gelatin silver print, 101×101 cm. Right: Soft robotics manipulator by Marchese et al. 2014. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

Biomorphic forms draw upon the organic shapes in plants or animals and are free-flowing, rather than structured according to a grid or rectilinear compositions as, e.g., geometric abstraction. These are, as is well known, abundant in surrealist visual art and sculpture. An apt example is Hans Bellmer's photographic series featuring the life size pubescent doll he constructed in the early 1930s (Fig. 1). A parallel to soft robotics is evident here, as similar bulging, bulbous, biomorphic forms are generated when

flexible fluidic soft robotic actuators made from silicone elastomers are actuated with pressurized air (see Fig. 1). Interestingly, in "Formless a Users Guide" Rosalind Krass in fact describes the uncanniness of Bellmer's series as related to the fact that, through the framing, staging, and reconfiguration of the doll that occurs in the photos, this life-like doll is reduced to "nothing but the swells and bulges of a pneumatic torso" [2:196], hinting at its at once highly organic and object-like character.

The surrealists' preference for biomorphic forms also resonates with their fascination of exotic life forms and animals in general and here lies another point of contact. Soft robots are frequently based on bioinspired design strategies, and moreover, their designs are usually derived from the physiology of bodies and limbs of fully or partially soft animals that are particularly apt for accomplishing certain mechanical tasks that roboticists aim to replicate [17]. André Breton wrote retrospectively in the essay "Caught in the Act" (1949) about the surrealist fascination with animals that: "The surrealist bestiary, gives pride of place, above all other species, to animals that are sui generis and have an aberrant or decadent appearance such as the platypus, the praying mantis, or the anteater" [3:129]. Similarly, the kinds of animals that have served as inspirations for soft robot designs are not common domesticated pets, but rather animals with which most humans have no sustained direct contact. These creatures possess radically othered bodies compared to the human one and include classes and species such as cephalopods, earthworms, caterpillars, jellyfish, and snails. In both surrealist art and soft robotics designs, we encounter appropriated organic shapes, structures, and proportions that can be traced back to outlandish appearing animal body parts from such species (Fig. 2).

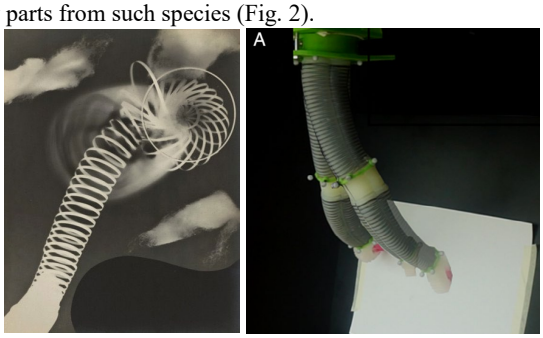


Figure 2: Left: Left: Man Ray, *Untitled Rayograph* (from the portfolio *Les Champs Délicieux*), 1922. Right: Soft robotic manipulator by Fischer et al. 2023. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

4 Material Transformation – Formlessness and Decay

There are also pictorial elements within surrealist visual art with which soft robotics share a connection not based on formalism nor referent but instead via a thematic interest in material transformations and the tension between structured organization and chaotic formlessness. The process of liquefaction or going shapeless, for instance, as exemplified in, among others, Salvador Dalí's famous soft watches or melting clocks in *The Persistence of Memory* (1931) (Fig. 3).

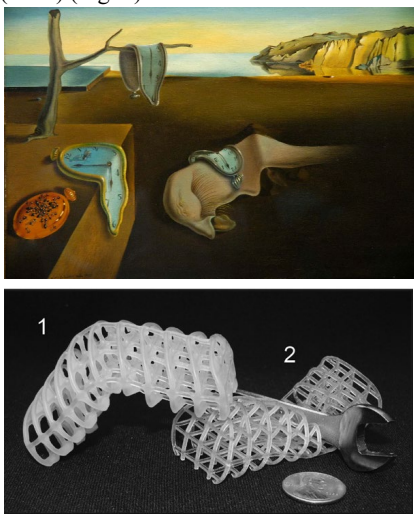


Figure 3: Top: Salvador Dalí, *The Persistence of Memory*, 1931. Oil on canvas 24.1 x 33 cm. Bottom: Dynamically softening lattice by Cheng et al. 2014. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

Within soft robotics research, this ability to change from being hard to being soft (and sometimes back again) has been pursued through several different technical means with a view to attaining robots that can deform their bodies to maneuver confined spaces or can adjust their mechanical firmness (impedance) for increased safety in collisions [20]. Fig. 3 (bottom image) shows an early example of this, where a wax lattice, imagined to constitute the body of a soft robot, was heated to switch between its solid (1) and soft state (2) [4].

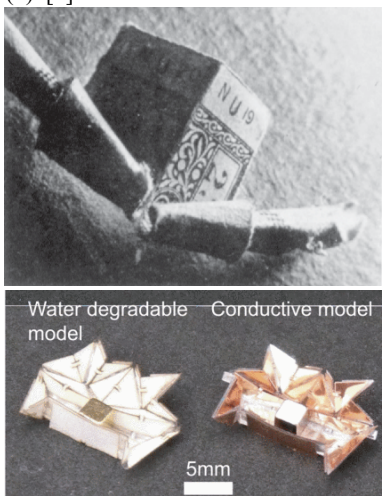


Figure 4: Top: Brassaï, photograph for Salvador Dalí *Involuntary Sculptures* (*Sculptures Involontaires*) (detail). Published in *Minotaure* (1933). Bottom: Soft origami robots

by Miyashita et al. 2015. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

Extending the theme of transitory existence implied in *The Persistence of Memory*, it is also worth noting how cheap, mundane, and mass-produced materials as well as matter capable of degradation fascinated the surrealists. In Brassaï's photographs for Dalí's series of involuntary sculptures, published in the dissident surrealist journal *Minotaure*, for instance, one sees pieces of paper scrap including bus tickets and theater stubs, which have had structures unconsciously imposed on them through fiddling, rolling, and folding by restless hands (Fig. 4 Top). These bear strong likenesses with soft robots and folded soft actuators made from paper and other degradable materials using origami techniques (Fig. 4 Bottom).

The involuntary sculptures photo series also includes a close-up photograph of a smear of toothpaste that recalls soft robots made from gooey materials such as deformable magnetic slime (Fig. 5).

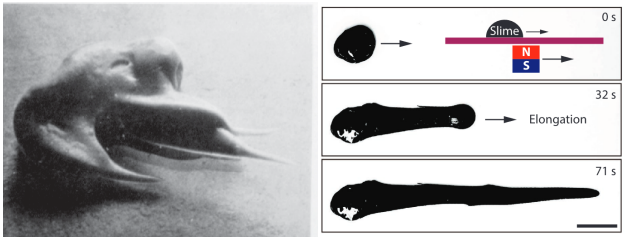


Figure 5: Left: Brassaï, photograph for Salvador Dali *Involuntary Sculptures* (*Sculptures Involontaires* (detail). Published in *Minotaure* (1933). Right: Soft magnetically-actuated slime robot by Sun et al. 2022. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

Such examples of unruly material accumulations of course also resonate with Georges Bataille's concept of the *informe*, the formless, which figures centrally in the work of the dissident surrealist group centered around Bataille. Bataille's informe constitutes "the category that would allow all categories to be unthought", as Rosalind Krauss puts it. The informe does not possess meaning but is posited by Bataille as having a task or a job – "to undo formal categories, to deny that each thing has its 'proper' form, to imagine meaning as gone shapeless". Hence, it "transgresses boundaries of terms, "producing formlessness through deliquescence, putrifaction, decay" [19:39–40]. Interestingly, here the surrealist aesthetic again arrives back at connection point with soft robotics research, which by now has a whole subfield of dedicated to constructing biodegradable robots from materials such as chitosan, gelatine, or cellulose that eventually can decay and be absorbed into natural environments more easily than rigid and traditional materials used for making robots [9].

5 Proto-Signification Through *Spacing*

The overlaps between surrealist aesthetics and the emergent aesthetic of soft robotics, I have highlighted so far, were identified via a strategy of close observations of form and content of surrealist artworks and by paying attention to the shared thematic interests of both surrealism and soft robotic research related to material process. However, within Rosalind Krauss' poststructuralist-deconstructivist theorization of surrealist photography, a parallel emerges that instead goes via structure and semiotic image operations. In the seminal article "The Photographic Conditions of Surrealism" Krauss argues that surrealist photography is characterized by a tension between image and physical reality, instituted by an infiltration of the photographic image by signification. Surrealist photographs, she asserts then "are not interpretations of reality…They are presentations of that very reality as configured, or coded, or written" [18:29]. Krauss arrives at this conclusion by theorizing the framing and visual manipulations occurring in surrealist photography as instances of what she, drawing on Jacques Derrida, refers to as spacing. Her argument is that spacing occurs in surrealist photography, through the frequent use of manipulations and techniques such as solarization, which accentuates contours as in, e.g., Man Ray's Lillies (Fig. 6), and doubling of the same motif within the photographic print.



Figure 6: Left: Man Ray, *Calla Lilies*, 1930, gelatin silver print, 28.58 x 23.5 cm. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

These photographic devices maintain the seamless unity of the photograph, but at the same time break it up from within, by compartmentalizing or repeating its elements so they are read sequentially rather than all at once (see Fig. 6) [18:22–25]. Through the repetition of a series of elements, Krauss argues, drawing on Claude Lévi-Strauss, a precondition for deliberate meaning is established, whereby "raw matter" is transmuted "into the conventionalized form of the signifier" and the paradox of seeing "reality constituted as a sign" emerges [18:25–31].

Albeit Krauss' theoretical account, concerns surrealist photography specifically, I want to argue that it can also elucidate the simultaneously artificial (i.e., constructed) and biological appearance of the large portion of soft robots that use a modular

and bioinspired design. Consider the two examples in Fig. 7 – a tentacle-inspired manipulator (top image) and an earthworm inspired crawling robot (bottom image). These robots are highly abstracted designs that visually diverge quite a bit from their biological inspirations or counterparts. But more importantly, they are designed and constructed by assembling a series of identical actuated modules to compose their bodies. Spacing of these parts is further augmented via the inclusion of passive elements, connector pieces or "spacers", that functionally only serve the purpose of holding together the actuated main parts of the morphology. Taken together, these aspects contribute to instituting repetition and a compartmentalization of parts within the coherent overall body plan of the robot, which echoes the photographic effect obtained through solarization or doubling within the photographic image and results in an infiltration by the sign, as described by Krauss.

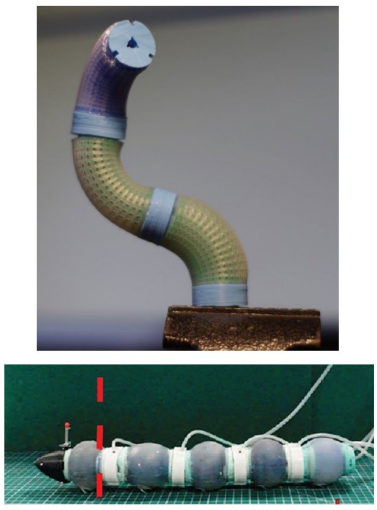


Figure 7: Top: soft robotics manipulator by Fras et al. 2020. Bottom: earthworm inspired soft robot by Das et al. 2023. DO NOT PUBLISH IMAGE - PERMISSION MUST BE OBTAINED

6 Soft Robots Will Be Convulsive?

In the preceding sections, I have sought to highlight parallels between the two historically separate and, in many respects, disparate phenomena of surrealism and soft robotics. The former a heterogenous revolutionary historical avant-garde art movement from the first half of the 20th century with a strong anti-rationalist bent, the latter an emerging technology with an associated set of techno-epistemic research practices anchored in positivist science, set a century later. I have argued that a surrealist aesthetic can be seen to resurface in soft robotics, despite no deliberate or explicit influence between the two. Yet it remains an open question what significance and effects the surrealist traits of soft robotics may have?

If one turns to empirical studies of how people interact with soft robots, published within the interdisciplinary research field of Human-Robot Interaction (HRI) there are indications that the contemporary aesthetic reception of soft robots indeed entails responses that dovetail with the experience of a central surrealist aesthetic concept, namely *convulsive beauty*. Convulsive beauty designates a combination of attraction and repulsion at the center of the surrealist notion of beauty and surrealist aesthetics. In fact, Krauss argues that "If we are to generalize the aesthetic of surrealism, the concept of Convulsive Beauty is at the core of that aesthetic". Convulsive beauty is both shocking and enchanting as well as unexpected and can arise from juxtapositions of elements that are disparate or contradicting, as in the famous quote by Lautréaumont (Isidore Ducasse), favored by the Surrealists, about the beauty of "the chance encounter of a sewing machine and an umbrella on an operating table" in Les Chants de Maldoror (1868). The concept first appears in Breton's novel Nadja (1928) and is further developed in his L'Amour fou (1937), which contains a part that was previously published under the title "Beauty Will Be Convulsive...." [18]. In the latter text Breton characterizes convulsive beauty through three examples, and the first of these interestingly center on mimicry of nature, which, as already mentioned, is also integral to the design practices of soft robotics. Moreover, in human-robot interaction studies on soft robotics technology, descriptions of experiences that align with the ambiguous aesthetic experience of convulsive beauty interestingly recur. In a study where people were asked to interact freely with robotic manipulators based on tentacle-inspired designs, for instance, this was the case especially with respect to how people experienced touching and being touched by the robots. As noted in the publication, "participants were intrigued to touch the robot, but also expressed ambivalence", and "For most participants, being touched by the robot seemed a somewhat transgressive yet simultaneously enjoyable experience" [15]. Similarly, in another recent study, where participants were exposed to prototypes of soft robots with designs that deliberately sought to amplify their biomorphic aspects, it was found that "[Their] organic qualities were variously … described as being desirable and uncanny" and that "Some participants .. felt that these qualities could contribute a sense of emotional comfort", while "Other participants … found … [them] … uncanny because they made the prototypes appear very 'natural-appearing' (P1.2) or like 'alien' versions of organic things" [5].

The dynamics encapsulated in the above quotations should of course not be taken to constitute a naturalizing nor deterministic aesthetics that is universally applicable, as the cultural and historically situated character of aesthetic processes cannot be elided. Hence, it is an open question to which extent culture modulates the experiences cited above that emphasize ambiguity and alterity in the encounter with soft robots, but also to what extent these qualities result from a novelty effect that will eventually wear off. The argument could perhaps even be made, that the fact that the study participants are able to make a distinction between at once both appealing and off-putting

qualities in soft robotics, might in part be a consequence of the surrealist aesthetic and sensibility's recuperation and diluted afterlife within popular culture. This would appear to indicate that time might have caught up with the surrealist aesthetic and robbed it off its once revolutionary potential. That is, within surrealist theory and practice, art assumed its importance because it could lead to liberation and revolutionize society through altering the individual's mind state and "normal" way of experiencing the world, by inducing the indistinction between imagination and reality characteristic of the experience of surreality. The testimonies of study participants, however, illustrate that there might be a potential and some distinct advantages in using surrealist concepts including convulsive beauty within HRI research on soft robotics, instead of the concept of "the uncanny valley" that is habitually cited by robotics researchers. This conceptual shift would, for instance, allow for articulating the coexistence of alluring and off-putting sensations in the aesthetic reception of artificial yet biologically appearing agents. Judging from the above, this appears to be a more adequate and nuanced way of conceptualizing experiences rather than referring to increasing or decreasing levels of affinity, as proposed by the "uncanny valley" theory [22].

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